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Authors: Jahan Heidari[a*], Jürgen Beckmann[b], Maurizio Bertollo[c], Michel Brink[d], Wolfgang Kallus[e], Claudio Robazza[c], and Michael Kellmann[a,f]

Affiliations: [a]Unit of Sport Psychology, Faculty of Sport Science, Ruhr University Bochum, Germany. [b]Technische Universität München, Department of Sport and Health Sciences, Munich, Germany. [c]University G. d'Annunzio of Chieti-Pescara, Department of Medicine and Aging Sciences, Chieti, Italy. [d]University of Groningen, Center for Human Movement Sciences, University Medical Center Groningen, The Netherlands. [e]Karl-Franzens-Universität Graz, Unit of Work, Organisational, and Environmental Psychology, Graz, Austria. [f]School of Human Movement and Nutrition Sciences, The University of Queensland, Australia, St Lucia, QLD, Australia.

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Title page

Full title of invited Review: Multidimensional monitoring of recovery status and implications for performance

Jahan Heidari[a*], Jürgen Beckmann[b], Maurizio Bertollo[c], Michel Brink[d], Wolfgang Kallus[e], Claudio Robazza[c], & Michael Kellmann[a,f]

[a]Unit of Sport Psychology, Faculty of Sport Science, Ruhr University Bochum, Germany
Postal address: Gesundheitscampus Nord Nr. 10, 44801 Bochum, Germany

[b]Technische Universität München, Department of Sport and Health Sciences
Postal address: Georg-Brauchle-Ring 60, 80992 Munich, Germany

[c]University G. d'Annunzio of Chieti-Pescara, Department of Medicine and Aging Sciences,
Postal address: Via dei Vestini, 31, 66100 Chieti, Italy

[d]University of Groningen, Center for Human Movement Sciences, University Medical Center Groningen
Postal address: A. Deusinglaan 1, 9713 AV Groningen, The Netherlands

[e]Karl-Franzens-Universität Graz, Unit of Work, Organisational, and Environmental Psychology
Postal address: Universitätsplatz 3, 8010 Graz, Austria

[f]School of Human Movement and Nutrition Sciences, The University of Queensland, Australia
Postal address: Blair Drive, St Lucia QLD 4072, Australia

Email addresses (in order of authors):
jahan.heidari@rub.de, juergen.beckmann@tum.de, m.bertollo@unich.it, m.s.brink@umcg.nl, wolfgang.kallus@uni-graz.at, c.robazza@unich.it, michael.kellmann@rub.de

[*]Corresponding author:
Jahan Heidari
E-mail: jahan.heidari@rub.de
Phone: +49 234 32 25981

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Abstract

Monitoring of recovery in the context of athletic performance has gained significant importance during recent years. As a systematic process of data collection and evaluation, the monitoring of recovery can be implemented for various purposes. It may aid to prevent negative outcomes of training or competition, such as underrecovery, overtraining, or injuries. Further, it aims at establishing routines and strategies necessary to guarantee athletes' readiness for performance by restoring their depleted resources. Comprehensive monitoring of recovery ideally encompasses a multidimensional approach, thereby considering biological, psychological, and social monitoring methods. From a biological perspective, physiological (e.g., cardiac parameters), biochemical (e.g., creatine kinase), hormonal (e.g., salivary cortisol) and immunological (e.g., immunoglobulin A) markers can be taken into account to operationalize training loads and recovery needs. Psychological approaches suggest the application of validated and reliable psychometric questionnaires (e.g., Recovery-Stress Questionnaire for Athletes) to measure a subjective perception of recovery as well as the subjective degree of training- or competition-induced fatigue. Social aspects also play a role in performance monitoring and may hence provide essential performance-related information. The implementation of a monitoring routine within athletic environments represents a continuous process which functions as an effective addition to training and depends on a range of conditions (e.g., organizational regulations, commitment of athletes). Current research in the field of monitoring aims at establishing individualized monitoring regimes that are referring to intraindividual reference values with the help of innovative technological devices.

Keywords: athletes, biological, psychological, review, health

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1. Introduction

There is a fine line between success and failure in elite sports. A plethora of considerable internal and external demands affects elite athletes and determines their performance in a positive or negative way.[1] Knowledge about potential stressors (e.g., expectations, injuries) and resources (e.g., social support, training facilities) in the respective sport environment can therefore serve as a valuable asset to prevent deleterious outcomes in terms of psychological and physical disorders (e.g., depression, musculoskeletal disorders).[2,3] The systematic collection of this information can be realized by sport scientists via the application of monitoring routines. By this means, elite athletes can be observed in training and competition settings with the aim of improving performance and maintaining well-being.[4]

Since the cohort of elite athletes is evaluated through its accomplishments, it is commonly surrounded by a team of professionals (e.g., coaches, physical therapists, sport psychologists, sport physicians) dedicated to maximizing performance. The concept of performance can be considered as the sum of behaviors individuals or teams show in order to reach specified standards and goals. It is influenced by the acquired physiological and psychological skill set and the reliable demonstration of these competencies under all circumstances.[5] Additionally, performance in sport was considerably increased over the years due to significant improvements in scientific research regarding performance-related aspects such as training, nutrition, technology, coaching, or psychology.[6,7]

The information obtained from all these facets of performance needs to be routinely collected and appraised to further optimize the skills and abilities of team or individual athletes. This systematic process of gathering and interpreting information is labelled as monitoring and can be applied for various reasons. Monitoring may serve the purpose of assessing whether an athlete is adapting and responding to a training program in terms of performance markers such as strength or endurance.[6] Monitoring may also aid to determine training and competition loads

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particularly in disciplines with intense competition schedules. Differences in biological (e.g., heart rate variability), psychological (e.g., motivation), as well as behavioral (e.g., GPS) parameters within and between athletes can be identified and appropriate adjustments to fatigue can be planned subsequently.[6,8]

A major field of monitoring does not aim at a direct performance enhancement. Rather, training load and subsequent fatigue is assessed via monitoring and can be used to quantify the recovery status (e.g., high vs. low recovered) in athletes. Based on these observations, recovery-related processes (e.g., sleep, nutrition) and activities (e.g., stretching, meditation) can be initiated during both training and competition.[9,10] As a set of complex processes in time, recovery takes place physiologically and psychologically with the aim of regaining a balanced psychophysiological state.[11] Since the need for recovery varies inter- and intraindividually and depends on internal and external influences, the monitoring of recovery requires a well-defined and structured framework.[12,13] Recovery does not only play a pivotal role in the restoration of lost resources and fatigued states to guarantee readiness for performance.[7,8] It is also important in minimizing the risk of negative outcomes such as overtraining, injuries,[14,15] and psychological disorders (e.g., depression[2]). The importance of monitoring recovery can best be explained by its relationship with performance and the negative consequences of too much training without appropriate recovery. Continuous non-functional overreaching (NFO) together with an emerging state of underrecovery characterize a development of decreasing performance and well-being. While NFO and underrecovery share many commonalities, they should be considered as distinct concepts. NFO refers to negative hormonal and psychological changes specifically related to training and competition contexts.[16] In contrast, underrecovery describes a broader condition of insufficient recovery also related to general stress (e.g., through friends, partner) independent from athletic activity.[17] Incipient NFO and underrecovery symptoms may manifest in psychological disturbances and hormonal

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perturbations that in turn are resulting in reduced performance of affected athletes.[16] If this downward spiral of excessive training, potential psychological and social stress, and insufficient recovery is not identified and stopped early enough, a state of overtraining may manifest as an ultimate consequence. High-level performance is no longer possible, while affected athletes experience severe psychological (e.g., amotivation, irritability, anger) and physiological (e.g., immunosuppression, cardiac disturbances) symptoms.[16]

An alleviation of these symptoms can be achieved through a combination of a significantly reduced training volume and frequency together with the application of long-term recovery strategies (e.g., systematic use of relaxation techniques). Depending on the individual severity of symptoms, even the prolonged cessation of athletic activity may be required.[18] To avoid such negative outcomes on performance and health, psychophysiological monitoring is increasingly gaining importance and popularity in elite and professional sports. It should be noted that recovery-stress monitoring needs to be conducted on a multidimensional, psychophysiological level to register subtle performance-related changes at the right time so that person- and situation-specific interventions can be launched.[17]

An ideal comprehensive monitoring encompasses psychological, social, and physiological factors.[19] Physiological monitoring comprises biochemical markers (e.g., cortisol, blood lactate) and electrophysiological indices (e.g., heart rate variability, individual alpha peak frequency), while psychological monitoring is mainly conducted with appropriate assessment tools (e.g, Recovery-Stress Questionnaire for Athletes[11]). Social monitoring includes family issues and the coach-athlete relationship.[6,18] Behavioral and performance indicators to quantify external loads also offer valuable information but will not be scrutinized in the context of this review.

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2. Biological approaches for monitoring recovery in athletes

Together with performance-based tests, biological parameters are labelled as objective indicators of fatigue in sports contexts. As a consequence of an intense training session, biological indicators of fatigue manifest in athletes and indicate the need of recovery to restore the depleted resources. Within athlete monitoring systems,[13] a number of biological parameters can be applied, ranging from physiological, biochemical, hormonal to immunological markers.[20] Due to the abundance of different available tools, we only present a selected number of strategies to provide a fundamental idea of biological monitoring possibilities. These biological measures are widely employed in sports settings and demonstrate a reasonable feasibility from a financial and temporal perspective. Physiological processes (e.g., derivations in cardiac indicators) resulting from training load and volume can be assessed to quantify the recovery status of the respective athlete.

As a non-invasive and practical instrument for the applied environment, reliable on-field heart rate recorders permit a continuous recording of cardiac variables over long periods (24 h or more), thereby including training sessions and subsequent recovery periods. Computer analyses allow the visualization of cardiac activity via a profile and indicate important values such as exercise heart rate (HR), heart rate variability (HRV; the specific change in time between consecutive heart beats), or heart rate recovery (HRR; the speed at which the heart returns to its normal rate after exercising).[8] Several authors suggest a combination of five minutes of resting HRV together with submaximal exercise HR (final minute of a 4–5 minutes lasting training bout) since these variables have shown to be the most feasible and useful cardiac parameters.[21,22] Research showed that measuring the resting HRV is easily conducted and delivers valuable output on both acute and chronic responses to training, while mirroring both positive and negative adaptations. General statements on how resting HRV specifically alters in response to underrecovery are difficult to provide due to a number of influential

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variables (e.g., individual vs. team sports, environmental conditions). Research should therefore be directed towards a domain specific (e.g., specifically for tennis) examination of cardiac variables and their alterations in light of overtraining. Individualized approaches to gather information about cardiac reactions may additionally enhance the meaningfulness of recommendations and interventions. On a general level, it can only be assumed that resting HRV is impaired in terms of higher HR in general together with a lower resting HR. This may occur as a result of a continuous activation of the sympathetic autonomous nervous system (ANS) reflecting a chronic stress response and psychophysical overload.[23,24] Integrating resting HRV may serve as one important part of monitoring the recovery status in case it is assessed over a longer period of time.[25] Identically, the importance of monitoring HRR for fatigue and performance alterations has also been proposed, but requires a few extra minutes of recording. It can be stated that signs of underrecovery as a precursor of overtraining include an increased resting heart rate, higher blood pressure, and an altered adaptation (i.e., slower recovery after exercise) of the heart rate to exercise together with derivations in the electrocardiogram.[20] However, all cardiac parameters need to be interpreted with caution and in accordance with certain guidelines.[6] Results should be related to the current training phase and load the athlete has to deal with as well as the individual development of these cardiac parameters. Measurement errors should be considered to assess the smallest meaningful change in these parameters in order to derive potential interventions.[6,21,26] For all cardiac parameters, it is essential to check for factors such as hydration, circadian rhythm, environment, and medication in order to exclude any potential confounders. The presented cardiac parameters exemplify feasible indicators of fatigue which have to be interpreted in relation to other markers of recovery status and gathered reference values of the respective athlete. They display cardiac alterations (e.g., lower HRV) for athletes with performance decrements who are at risk of underrecovery and overtraining.[20]

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The aim of biochemical monitoring consists in the identification of metabolic processes and residues of muscle damage related to performance and training responses. They are mainly assessed via blood samples, urine, and saliva and show proneness to interindividual variability.[6] Creatine kinase (CK) is the most frequently used biochemical marker across a plethora of sport disciplines and is therefore discussed within this review.[8,20] CK provides information on muscle damage and is supported by a significant body of research in the field of training load research for sport disciplines which focus on muscular and mechanical loads. It appears that intense exercise causes damages of the skeletal muscle cells and is associated with increased total serum CK.[27] However, an indication of overtraining based on certain levels of CK cannot be inferred as a general standard, especially for sport disciplines without an accentuated physical profile (e.g., shooting). CK exemplifies another biochemical marker which should be evaluated in strength dominant sports within an individualized framework when compared to standardized personal values.[7]

Hormonal changes have been investigated in the context of athletic training and monitoring of the impact of training on organismic functioning and illness in athletes. Salivary cortisol and testosterone have been recommended to be combined into a testosterone/cortisol (T/C) ratio. As a guideline, tapering off is suggested in case the ratio slightly lowers, while overtraining may occur if the ratio dropped by more than 30%.[28] Caveats should be noted for certain sport disciplines such as cycling since testosterone is affected by the compression of the saddle on the gonads. Moreover, the relationship between performance and these indicators remains weak and a long-term follow-up of an athlete is required for reasonable conclusions.[29] The value of these measures to determine the internal load regularly has not been studied and none of these hormonal factors has been identified as definite marker of fatigue.[8]

Immunological deficiencies represent a prominent herald of overtraining through an increased frequency in the occurrence of colds and unresolved viral infections.[20] These

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weaknesses can be detected via collecting blood samples (low leucocytes and rise in lymphocytes). Moreover, secretory immunoglobulin A delineates another immunological marker of augmented fatigue and can be assayed via saliva samples. Both methods characterize feasible immunological strategies that can be used to detect fatigued states in athletes.[20] While all biological measures certainly contribute to a better understanding of training induced fatigue and performance related responses to training loads, no single biological marker can be applied as gold standard of monitoring. The decisions for a specific marker depend on a number of factors, for instance the timeframe, financial resources, and trained personal within the athletic staff.[13]

3. Psychological approaches for monitoring recovery in athletes

A different approach to operationalize responses to training loads in athletes targets the measurement of psychological consequences of training-induced fatigue to indicate the recovery status as a basis to determine and initiate recovery activities (e.g., relaxation techniques). Psychological monitoring instruments are commonly referred to as subjective self-report measures since they reflect the individually perceived recovery status of the respective athlete. These instruments aim at the assessment of changes in psychological states such as mood and stress, but primarily encompass the measurement of recovery in reaction to acute or chronic training loads.[13] In light of continuous monitoring with psychological measures, gradual and subtle negative developments in health status (e.g., underrecovery, NFO) can be directly identified to prevent detrimental outcomes such as the overtraining syndrome. Sport-scientific research has produced a number of suitable and sensitive psychometric questionnaires which are based on scientifically developed theoretical models. The majority of these instruments are currently used in sport-practical contexts, thereby emphasizing a promising transfer from theory to practice. Research suggests that the established measures vary regarding their purpose and target group. Therefore, they should be carefully selected

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based on the paramount trainings or competition objective of the respective team or individual athlete (e.g., injury prevention, performance enhancement).[13]

Some measures evaluate well-being on a general level, while others demonstrate clear sport-specific item phraseology. Self-report measures can also be differentiated based on their assessment of single (e.g., stress) or multiple constructs (e.g., vigour, anger). However, a comprehensive measurement of the psychological state of an individual is indicated in the majority of cases.[30] Some instruments focus on the assessment of symptoms to obtain a clinical overview of physical (e.g., headache) or behavioral (e.g., social withdrawal) repercussions of training.[13] From a practical point of view, a preventive approach of monitoring alterations in mood, stress and recovery prior to the manifestation of symptoms appears to be the most beneficial.[12] To provide a better idea of psychological monitoring in sports, we briefly present three of the most commonly used instruments in practice,[13] namely the Multi-component Training Distress Scale (MTDS[31]), the Profile of Mood States (POMS[32]), and the Recovery-Stress Questionnaire for Athletes (RESTQ-Sport[11]). A more comprehensive overview and categorization of psychological assessment tools is provided by Heidari et al.[18]

The MTDS represents a multidimensional self-report measure which assesses a variety of different psycho-behavioral aspects. These facets are organized in three key domains (stress, mood, physical/behavioral symptoms) in reference to intense and demanding training sessions (e.g., training camps). Six broader symptom clusters (*Depression*, *Vigour*, *Physical Symptoms*, *Sleep Disturbances*, *Perceived Stress*, and *Fatigue*) are documented by 22 items combining intensity scales to assess mood and physical/behavioral symptoms with a frequency scale to assess perceived stress. The compactness of the MTDS qualifies the instrument as a cost-efficient and informative instrument for both practitioners and athletes in various sports contexts.[31] Another frequently used instrument suitable for the identification of mood states in athletes is the POMS. As a pioneer psychometric tool in overtraining research, plenty of studies

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highlight the relevance of the POMS as a monitoring instrument. It comprises 65 items categorized in five negative affect scales (*Tension*, *Depression*, *Anger*, *Fatigue*, and *Confusion*) and one positive affect scale (*Vigour*). A POMS global score can be generated by subtracting the *Vigour* score from the sum of the five negative scales while adding a constant of 100 (*Tension* + *Depression* + *Anger* + *Fatigue* + *Confusion* + 100 – *Vigour*). The POMS answer mode can be adapted from "How have you been feeling during the past week including today" to "How are you feeling right now" based on the primary aim of application.[33] Via the assessment of the six mood dimensions (i.e., *Tension*, *Depression*, *Anger*, *Vigour*, *Fatigue*, *Confusion*), an 'iceberg-profile' can be generated, with high values of *Vigour* being considered as favorable.[18] While the POMS represents a very common instrument used in sport settings, the Brunel Mood Scale (BRUMS[34,35]) as a shortened version of the POMS may serve as a suitable instrument for practitioners to be implemented in monitoring contexts. Both the POMS and the BRUMS characterize useful instruments to discover mood disturbances prior or after intensified training, in competition settings or within long-term monitoring approaches.[12,30] Both questionnaires entail specific recovery-related aspects (e.g., mood, stress) but should not be categorized as prototypical instruments to assess the recovery status with all its facets.

A comprehensive psychometric questionnaire to measure the recovery status is the RESTQ-Sport.[11] The instrument includes the thorough measurement of recovery and stress factors to obtain a detailed picture of the recovery-stress state of the respective athlete or team regarding a time frame of the previous three days and nights. Sport-specific as well as external factors (e.g., social activities) are recorded with a total of 19 scales, with four items describing each scale. The dimension Overall Recovery entails the scales *Success*, *Social Recovery*, *Physical Recovery*, *General Well-being*, and *Sleep Quality* as well as Sport-Specific Recovery with the scales *Being in Shape*, *Personal Accomplishment*, *Self-Efficacy*, and *Self-Regulation*. Accordingly, stress is divided into Overall Stress encompassing the scales *General Stress*,

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Emotional Stress, *Social Stress*, *Conflicts/Pressure*, *Fatigue*, *Lack of Energy*, and *Physical Complaints* while Sport-Specific Stress comprises *Disturbed Breaks*, *Emotional Exhaustion* and *Injury*. These scales can be illustrated via a recovery-stress profile to describe the balance between the two concepts. The RESTQ-Sport has been established as a monitoring instrument in sports contexts and can be applied as an indicator of (chronic) underrecovery and (chronic) non-functional overreaching including injury occurrence.[9,14,15] A recently developed, abbreviated version of the RESTQ-Sport with only 36 items and 12 scales has been introduced. In accordance to the longer version, the dimensions Overall Recovery (*Social Recovery, General Well-being, Sleep Quality*), Sport-Specific Recovery (*Being in Shape, Personal Accomplishment, Self-Efficacy*), Overall Stress (*General Stress, Social Stress, Fatigue*), and Sport-Specific Stress (*Disturbed Breaks, Emotional Exhaustion, Injury*) can be calculated as a mean of the respective scales. The RESTQ-Sport-36 may serve as a more user-friendly tool from a practical standpoint.[11]

4. Social approaches for monitoring recovery in athletes

Although the area of social monitoring appears to play a subordinate role in athletic contexts, several findings in current research address the importance of social factors (e.g., coaches, family, and teammates) for performance.[6,36] It can be suggested that these social facets impact important moderators in the monitoring context, such as compliance, susceptibility to stress, and dealing with recovery to avoid undesired performances.[37] A functioning social environment with supportive climate can serve as a strong protector against the adversities in sports settings and as a performance stimulant.[38] Clear monitoring guidelines for the social domain do not exist, but positive, appreciative communication with athletes in combination with the possibility to interact with family members and friends on a regular basis might contribute to well-being and continuous high-level performance.[39] Validated psychometric instruments which include social facets remain scarce in the sports context, but we propose an

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assessment tool that might be used as a social monitoring instrument. A systematic application of this questionnaire might help to diagnose social issues as obstacles to performance in athletes. The identified problems can then be addressed in conversations with individual athletes or teams.

The Daily Analysis of Life Demands for Athletes (DALDA[40]) describes a composite self-report measure with the aim of evaluating an individual's stress state and factors contributing to the current stress condition. It can be implemented to measure psychophysiological and social stress as a consequence of variations in training load.[41] It consists of 34 items and is divided into two segments. The first part (Part A) identifies nine sources of stress (e.g., home life, school/college/work) while the second part (Part B) targets the evaluation of 25 stressors (e.g., recovery time, irritability). Participants label the ratings for Part A and B as either 'normal', 'worse than normal' or 'better than normal' resulting in a straightforward and comprehensible answer mode. Subsequently, the source and symptom responses are portrayed graphically on a timeline. Athletes should complete the DALDA on multiple occasions over a pre-determined period of time to enable a comparison of the results with preceding training sessions. In case an augmenting number of 'worse than normal' responses are reported on three consecutive days, an unfavorable stress state might be present warranting for interventions and training modifications.[40] A strong advantage of applying the DALDA consists in the integration of both non-sport and sport-related sources and symptoms of stress. By including social aspects of stress, other important perspectives and origins of stress may be detected and discussed. The DALDA provides detailed insights about changes in social functioning and physical and behavioural responses to training. The scoring system and assessment method is very comprehensible, which facilitates the implementation in athletic contexts. Apart from the DALDA, the previously described RESTQ-Sport encompasses social functioning with three specific scales (*Social Stress*, *Conflicts/Pressure*, *Social Recovery*).

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Continuous deficits related to the social domain in the RESTQ-Sport might also act as warning signals for social conflicts which could be solved by seeking the talks with respective athletes. It can be stated that monitoring social processes should strongly be promoted within the athletic context and is ideally performed by integrating psychometric measures and direct communication. The direct social interaction in terms of personal communication in combination with questionnaire data may provide more detailed insights into the recovery status of affected athletes. Social recovery monitoring offers the possibility to detect facets of fatigue and recovery that are not covered by current biological and psychological monitoring approaches.

5. Bridging the gap: From theory to practice

Establishing a monitoring routine among athletes or teams depicts a gradual process requiring a number of prerequisites to serve as a successful addition to training. The commitment and agreement regarding the idea of monitoring should be obtained from stakeholders, coaches, athletes, and other staff members as they form the basis of the entire monitoring process.[12] Stakeholders should be informed about the purpose and aims of monitoring, and coaches should be aware that the conclusions derived from the monitoring data might potentially affect their training principles. It is crucial to consider the fine line between coaches' interpretation of monitoring as support or as a challenge of their coaching ability. Other authors with experience in the practical work with athletes add that the burden on athletes and the involved staff should be limited to a minimum to enhance compliance, data quality, and continuity in monitoring.[12,42] It should be noted that transparency regarding the monitoring procedure together with a theory-based and thorough explanation of the rationales fosters the commitment of all parties to be part of the monitoring process. A neat and practical decision tree which describes a sequence of different steps to be considered during the implementation of monitoring routines in athletic contexts has been recently proposed.[12] While the authors refer

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to subjective self-report measures, the basic principles can be applied to biological monitoring as well.

From a practical perspective, the benefits of monitoring for athletes need to be clarified in advance. They should consist of an individualized design of training routines to enhance performance and well-being.[42] In coordination with the athletic staff, a suitable monitoring instrument should be selected based on the previously specified goals (e.g., improvement of maximal strength). This decision also depends on aspects such as schedule, availability of time, financial resources, or other organizational aspects. For example, monitoring at isolated key points (preseason, midseason, postseason) or selected intervals (preseason - first competition) during a season could be conducted with more effort and investment in terms of financial and personal resources. Educated and experienced staff could engage in a combination of both psychological (e.g., RESTQ-Sport) and physiological (e.g., salivary cortisol) measurements to obtain multidimensional reference values for the recovery status. This *modus operandi* is indicated for the analysis of long-term developments or the comprehensive analysis of training camps or longer-lasting competitions. Ideally, a combination of key assessments at certain points and regular monitoring (e.g., weekly, monthly) with brief measures (e.g., MTDS) is recommended by Heidari et al.[18] and Saw et al.[12] This approach helps to develop targeted interventions for monitored athletes. With the extensive information gathered from assessments at the important time points, long-term interventions can be designed. In contrast, the regular monitoring with shorter instruments may serve as an immediate indicator of the recovery status resulting in acute interventions. Current research has recognized the requirements of sport-practical environments by fostering the development of short and comprehensive psychometric measures such as the Acute Recovery and Stress Scale (ARSS[43]) and the Short Recovery and Stress Scale (SRSS[43]). These two recently developed questionnaires allow the quantification of the current recovery-stress state. The ARSS

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comprises 32 adjectives related to both recovery and stress. Eight scales can be generated based on these adjectives resulting in each four recovery (e.g., *Mental Performance*) and stress scales (e.g., *Muscular Stress*). The shortened version of the ARSS is represented by the SRSS. The SRSS entails eight items (four recovery/four stress items), which are comprised versions of the eight scales of the ARSS.

Alterations of subjective loads and potential deviations from the regular perception of training or competitions can be detected by implementing these subjective procedures.[12]

As a consequence of a systematic monitoring cycle, a significant amount of data accumulates which needs to be evaluated appropriately. Independent from team or individual contexts, the main focus should lie on individual reactions to training as well as the individual appraisal of the own physical and mental condition. The quintessence of the entire process consists in the identification of practically meaningful developments and subsequent interventions. Therefore, recent research suggests the application of comprehensible and transparent statistics such as graphs and the visualization of the smallest meaningful change[26] which can then be communicated directly to the athletes and coaches.[42,44] The systematically collected data should be synthesized according to the knowledge and background of the target population of athletes. As a final and essential outcome of monitoring, training routines in team or individual contexts can be adapted and individualized recovery strategies can be implemented.[17]

For other applied areas, self-monitoring tools could also become more and more important. Monitoring of the recovery-stress state at work may yield valuable information to keep high-performance potential and to prevent negative outcomes in working contexts.[45] Suitable self-monitoring regimes need to be developed while modern technical possibilities may certainly contribute to accelerate and facilitate monitoring approaches. A proper

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application of self-monitoring in the working environment may help to identify problems and resources related to the workplace.[46]

6. Limitations and future directions

While monitoring and data collecting may significantly contribute to athletes' performance and health, it is also accompanied by a number of drawbacks that need to be considered prior to and during implementation. Data quality of subjective measures may suffer in case athletes do not comply with the scheduled dates or enter values that do not correspond to their actual condition. This issue needs to be addressed in the planning phase, as the gathered data may influence subsequent recommendations, interventions, and changes in the training structure. A trustful coach-athlete or sport psychologist-athlete relationship together with respectful communication of all parties increases the probability of obtaining meaningful data that can be transformed into reasonable and beneficial suggestions and outcomes.[47] With regard to monitoring approaches on a biological level, considerable costs and a time lag between field assessment and laboratory output are present for a number of measures (e.g., blood samples). These logistical issues contradict the applicability of these methods for daily monitoring in case the resources for these procedures are not provided. It also has to be noted that a consensus on the optimal or practically most relevant biological parameter has not been reached so far.[8,13] While overtraining has been examined from a biological point of view, findings on physiological, biochemical, hormonal, or immunological underpinnings remain inconsistent.[13]

Biological markers are prone to a considerable intra-assay and inter-assay variability as well as to intraindividual and interindividual variabilities that may depend on circadian rhythms, climate, sleep status, nutrition and hydration status, psychosocial factors and exercise routines. However, the difficulty of distinguishing normal physiological variations from symptoms of maladaptation as a result of heavy training loads remains.[13,48]

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Due to these limitations, state-of-the-art monitoring research suggests to engage in individualized approaches to acquire intraindividual reference values.[49] These can be related to comparable competitions or exercise bouts and can be established after a period of systematic data collection. An association between the right dosage of training in relation to performance can be established and may serve as a basis for changes in training.[6] Statistical approaches such as the smallest meaningful change can be used to assess noticeable deviations in both positive and negative directions. While the superiority of subjective monitoring instruments is assumed,[13] best practice should aim at a multimodal approach including biological and social aspects in the athletic context.[50] Each facet of monitoring (i.e., biological, psychological, social) provides specific insights about aspects of athletes' recovery status which cannot be gathered by exclusively relying on one specific approach. Figuratively speaking, the puzzle of an athlete's recovery status therefore requires biological, psychological, and social pieces to be considered complete. Finally, technological progress has led to the development of a number of monitoring devices, including wearable sensors, GPS trackers, and apps. These innovative methods to monitor athletes instantly gather information about training-specific underpinnings. Preferably, these techniques should be intuitive, small, and cost-effective while offering new perspectives for training design and evaluation. The application of specific monitoring apps for instance can harness a direct visualization of training phases immediately after a training session. In case a coach is not able to accompany the athlete personally, these digital outputs can nevertheless be discussed and analyzed via video contact. The steady progress and improvement of technology requires to stay updated and educated about the appropriate applicability of each device or method and the usefulness of the data in light of the overall goal of the respective athlete or team.[6] Possibilities to monitor athletes' performance and health are abundant; the challenge of practitioners in the field

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consists of the selection of an appropriate monitoring program in consultation with all individuals involved.

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